

# Hand Redirection Thresholds in a Shooting Game

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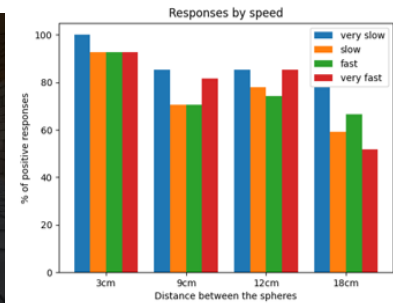
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(a) Real World Setup



(b) Virtual Environment



(c) Results of User Study

Figure 1: (a) A participant engaging in a shooting game, interacting with a physical proxy to switch weapons. (b) The virtual environment where the game occurs, showing virtual spheres on the table. Unbeknownst to the participant, their hand is redirected to touch the same haptic proxy, regardless of the virtual sphere they attempt to reach. (c) Histogram of manipulation detection rates collected during the user studies, comparing the two experimental conditions.

## ABSTRACT

This study investigates the combined effects of time pressure and interactivity on hand redirection perception in virtual reality during haptic retargeting, using an immersive shooting game as the experimental context. A total of 27 participants interacted with a haptic proxy under two experimental conditions: varying the speed of approaching enemies and the distance between the physical and virtual objects. The results revealed a high perceived normality of interactions, with over 50% of responses rated as normal across all phases of the experiment. These findings highlight the positive influence of engagement on the perceived realism of haptic interactions.

**Index Terms:** Hand Redirection, Haptic Retargeting, Perception.

## 1 INTRODUCTION

Advancements in head-mounted displays, such as improved field of view, frame rate, and tracking, have propelled VR toward more immersive experiences. Another critical aspect affecting immersion in VR is haptics. Virtual experiences often lack tactile feedback, meaning users cannot touch or feel virtual objects. While physical counterparts for virtual objects offer a solution, they are typically expensive, require precise tracking, and demand physical space to match their virtual counterparts. To address this limitation, Azmandian et al. [1] proposed a novel solution using a single proxy (physical object) that can represent multiple virtual objects. This approach relies on hand redirection, a technique inspired by redirected walking, where the movement mapping between the virtual hand and the physical hand is deliberately mismatched.

This method enables a single physical object, such as a sphere, to be used for interacting with multiple virtual objects. For instance,

the hand is redirected so that the virtual location of each of the three virtual sphere aligns with the physical sphere, even though their actual positions do not match. While this method offers a promising approach to enable haptics in VR, it needs to be ensured that the manipulation remains subtle to minimize cybersickness and maintain immersion by avoiding noticeable breaks in presence. To achieve this, thresholds must be established to define the level of mismatch between the virtual and physical object positions that users can tolerate without noticing. While such thresholds have been explored in the literature, they vary significantly based on factors such as the redirection method used, how the mismatch is measured, and the specific task users perform during the experiments. Most of these studies have been conducted in controlled environments where users are aware of the manipulation and perform simple tasks, such as touching objects, often with minimal distractions. This motivated the goal of our work: to examine the thresholds of hand redirection in a more realistic scenario — a shooting game — where users are engaged in a complex, immersive task. Additionally, unlike previous studies which often rely on controllers or specialized tracking hardware, this work investigates thresholds using hand detection based solely on Head-Mounted Display (HMD) tracking. This approach provides a more practical and accessible method for assessing redirection in real-world VR applications.

## 2 BACKGROUND

Hand redirection has been a significant research topic as it offers a feasible way to implement haptic feedback in VR applications, despite the limited availability of physical proxies in practical scenarios. Determining the thresholds for hand redirection is crucial to ensure realism and immersion. Various studies have explored this topic using different methods to measure these thresholds. Some have relied on incremental hand retraction, while Benda et al. [2] have focused on instantaneous hand displacement. Furthermore, the measurement of discrepancy varies between studies. Zenner and Krüger [4] expressed it as the distance difference between the final positions of the virtual and physical objects without considering direction or as a discrepancy defined in terms of angular offsets. Typically, thresholds are specified based on three directional components, resulting in three distinct thresholds. While various stud-

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ies present values for these thresholds, this work expands on that by focusing on the interaction task used while the hand redirection is applied.

### 3 METHODS

The study we conducted closely followed earlier studies involving the Three Cube Illusion [1]. We developed a dungeon-like VE that incorporated three types of enemies. Each enemy could be destroyed using a specific type of weapon. The user had to touch a sphere representing the corresponding weapon type to select it. One physical sphere was placed in the real-world space, which the user continuously touched to interact with the virtual spheres. The sphere was placed to the side of the user to force them to look around, lowering their focus time. At the start of the study, three physical spheres were placed to obscure the manipulation from the user. However, the two side spheres were removed as soon as the participant put on the HMD. It should be noted that before the user could touch the sphere they had to first touch a virtual cube that was used to start the redirection algorithm. This was done to ensure that each redirection had a common point of origin, such that the redirected hand movements are comparable to one another.

To perform the shooting action, the user rotated their wrist, releasing a ball of the selected weapon type in the direction they are looking. If the target was hit, the user could shoot again immediately. However, if the target was missed, the user had to wait 2 seconds before shooting again. The hand redirection was implemented using the HaRT toolkit [3]. The environment was streamed from a PC equipped with an Nvidia Titan RTX graphics card to an HTC Vive Focus 3. The physical space included a chair with a table to the left of the user, encouraging them to move and interact with the spheres. The experiment proceeded as follows: the user was introduced to the study without revealing the manipulation. They signed a consent form and completed an initial questionnaire, followed by a tutorial environment where they could shoot the three different types of enemies and interact with the spheres. Once the user was ready, the main experiment began. The experiment consisted of 16 rounds, each lasting one minute. In each round, multiple enemy types spawned consecutively, and the user had to switch weapons by touching first the cube and then the corresponding sphere. For the experiment, we varied two factors:

- The distance between the virtual and physical spheres, which was set at 3 cm, 9 cm, 12 cm, or 18 cm.
- The speed at which enemies spawn and walk, creating varying levels of time pressure. The speeds were categorized as very slow, slow, fast, and very fast.

The levels were counterbalanced, ensuring each user experienced all combinations of conditions. After each round, the question, “Did the interaction feel normal?” was answered. The response was a simple yes or no based on a two-alternative forced-choice approach.

### 4 RESULTS

Twenty-seven participants, aged 18 to 49 years, took part in the study. They were primarily recruited through a poster shared with students enrolled in specific virtual reality-related courses. Among the participants, 13 identified as female, 13 as male, and one as non-binary. No participant experienced a significant increase in the Simulator Sickness Questionnaire (SSQ) scores between the pre- and post-study questionnaires; therefore, no participants were excluded from the study. The responses to the two-alternative forced-choice question are shown in Figure 1c. The positive answers represent the percentage of participants reporting a normal interaction for the given levels of the conditions. Across the 16 phases of gaming, where all permutations of the two condition levels were

experienced, more than 50% of participants consistently reported the experiment as feeling normal, even when the mismatch distance reached 18 cm- a distance previous studies [2] rarely found found positive responses for. However, it should be noted that as the speed of the enemies increased, causing greater time pressure, the likelihood of participants reporting normalcy decreased significantly. This finding is supported by logistic regression analysis, where  $p < 0.05$ , indicating that higher enemy speeds negatively impacted the perception of normal interaction. Thus, it was concluded that time pressure negatively affects thresholds, whereas engagement in a game in general has a highly positive effect on them.

### 5 DISCUSSION AND CONCLUSION

Our study shows mixed results where increasing enemy speeds—and thus time pressure—did have a negative effect on the normal response of participants. This could be due to the particular set up of the experiment where the placement of the spheres to the side forced participants to look away from the threat of the approaching enemies whenever they needed to pick a new weapon. This indeed reduced their focus time, but besides increasing time pressure, this also resulted in more rapid head rotations. As we relied solely on the HMD tracking system, participants’ hands often moved out of the HMD’s field of view, resulting in a loss of tracking. This not only led to inaccurate hand position data but also impaired the redirection algorithm. The incremental displacements accumulated during the movement, and when the participant’s hand re-entered their field of view, these small increments appeared as a single large, instantaneous displacement. This made it significantly more likely for participants to detect the discrepancy. Future work could address these challenges by ensuring reliable hand tracking throughout the movement and employing more advanced redirection algorithms capable of mitigating sudden jumps and discrepancies. Despite these negative impacts on the detection rate of discrepancies, the rates have been overwhelmingly high over all phases: even during very fast enemy speeds and displacements of 18 cm. Pushing the boundaries of comparable work, we show the profound camouflaging effect that engagement has on redirection offsets. The results of this work extend previous literature on hand redirection perception thresholds, providing a novel application where they are tested in a gaming scenario, and this is hopefully a promising direction for future exploration.

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